De-Duplication Using Locality Sensitive Hashing

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Programming Practicum (CS 571)

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Scaling the Heights

IIT Mandi

Indian Institute of Technology Mandi

Problem Statement and Applications

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  - Recommend similar products, movies etc.

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- High dimensionality of the unique characteristic/feature vector required to identify documents.
- Documents can be very large or so many, may not possible to fit in main memory.
- In case of too many documents(say 1 billion) to compute pair wise similarity for each pair of document is highly expansive and time taking.

Big Picture of De-duplication Model

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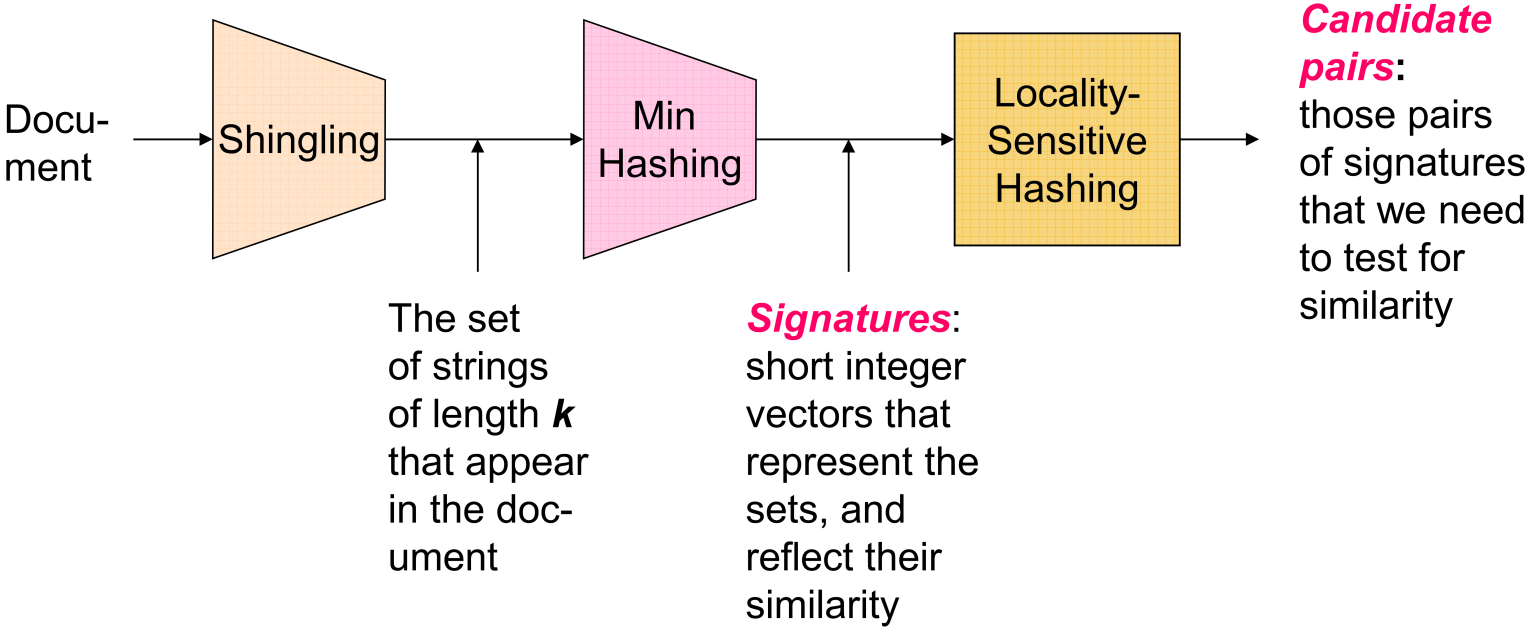

Source: J. Leskovec, A. Rajaraman, J. Ullman: Mining of Massive Datasets

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  $\boldsymbol{D1}$ **:** De duplication Using Locality Sensitive Hashing

  $\boldsymbol{D2}$ **:** Using Locality Sensitive Hashing for De duplication

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$$\boldsymbol{SIM(D1, D2) = \frac{|D1 \cap D2|}{|D1 \cup D2|} = \frac{4}{7} = 0.57}$$

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$D1 \cup D2$	$D1$	$D2$
“De duplication"	1	1
“duplication Using"	1	0
“Using Locality"	1	1
“Locality Sensitive"	1	1
“Sensitive Hashing"	1	1
“Hashing for"	0	1
“for De"	0	1

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- Suppose the rows of the boolean matrix permuted under random permutation $\boldsymbol{\pi}$
- Define a **hash** function $\boldsymbol{h_\pi(C)}$ = the index of the first (in the permuted order $\boldsymbol{\pi}$) row in which column $\boldsymbol{C}$ has value $\mathbf{1}$:

$$\boldsymbol{h_\pi(C) = min_\pi \pi(C)}$$

Example of Signature Matrix

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Permutation π | Input matrix (Shingles x Documents)

2	4	3	1	0	1	0
3	2	4	1	0	0	1
7	1	7	0	1	0	1
6	3	2	0	1	0	1
1	6	6	0	1	0	1
5	7	1	1	0	1	0
4	5	5	1	0	1	0

Signature matrix *M*

2	1	2	1
2	1	4	1
1	2	1	2

Similarities:

	1-3	2-4	1-2	3-4
Col/Col	0.75	0.75	0	0
Sig/Sig	0.67	1.00	0	0

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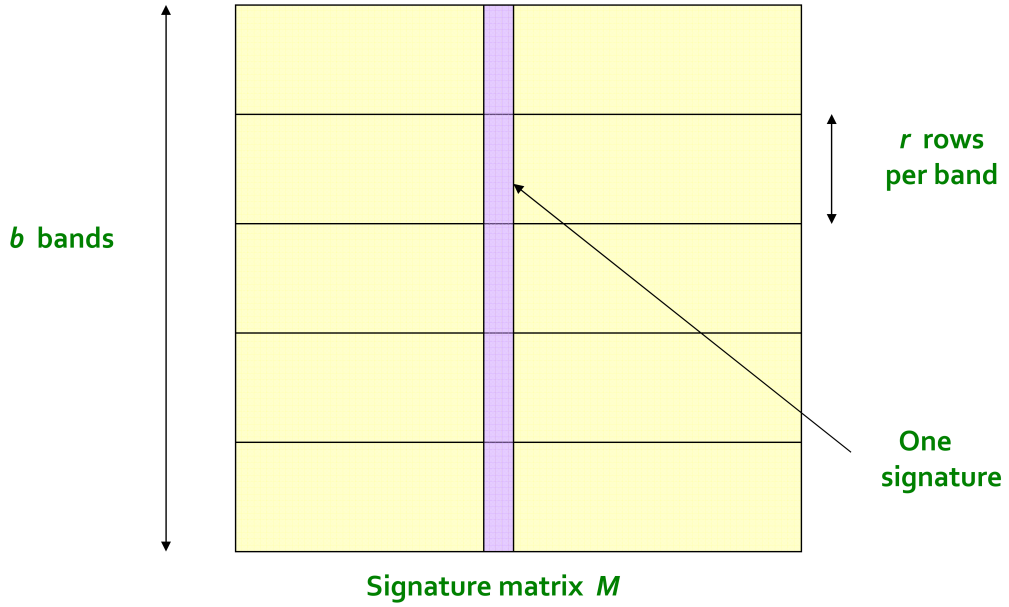

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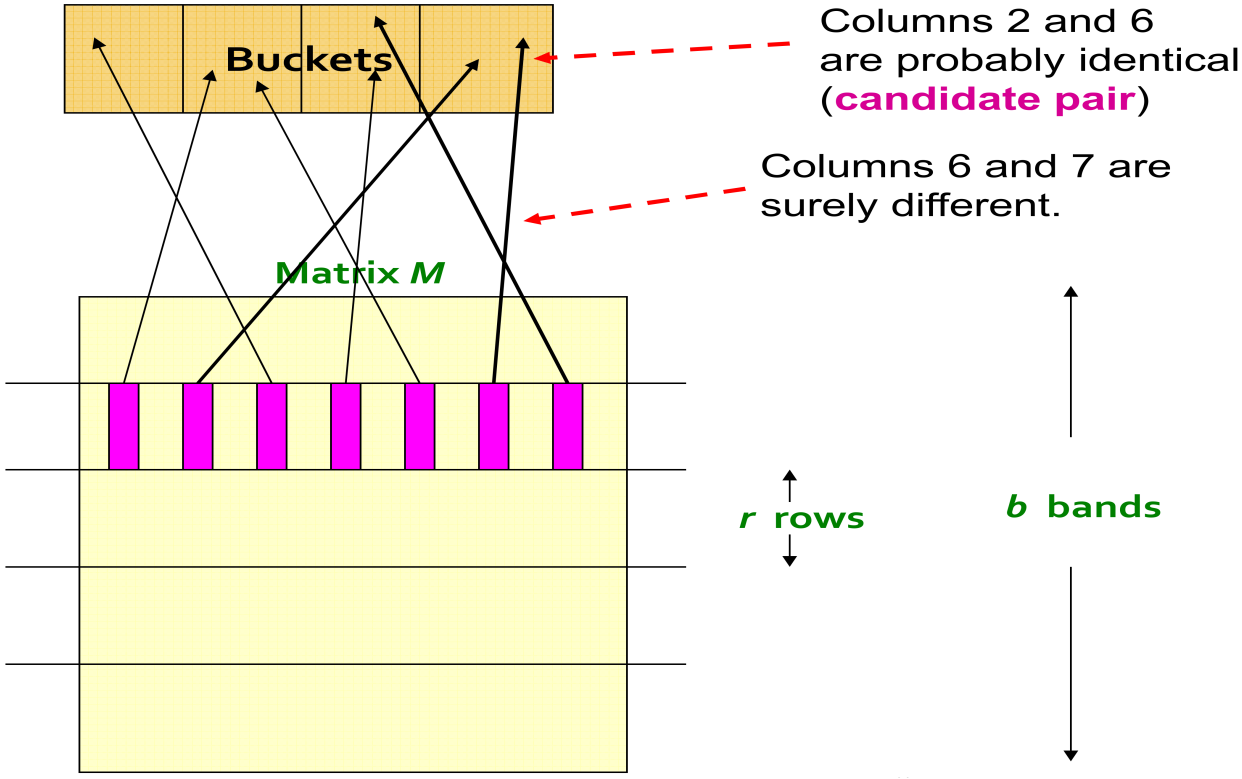

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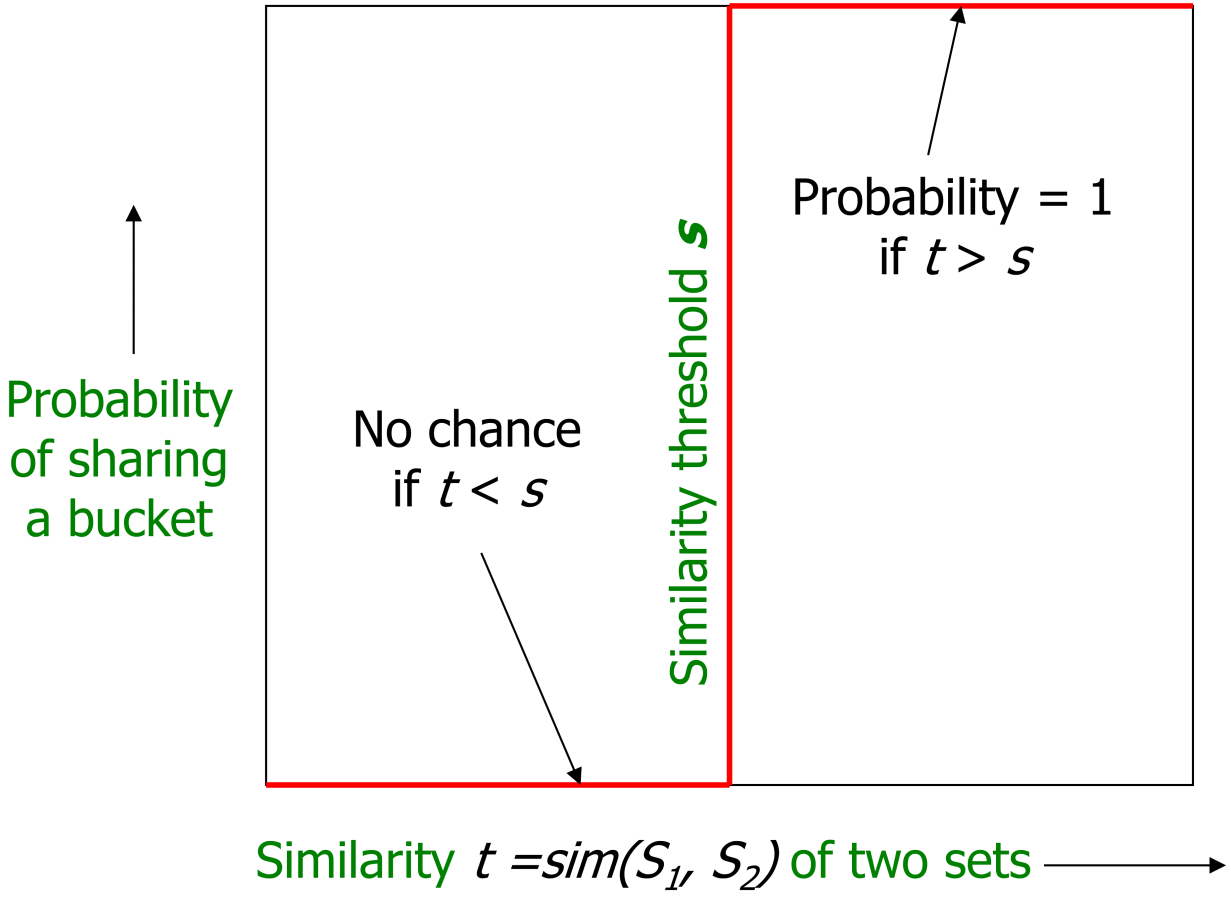
Interpretation of b bands for r rows:

- We want that for column whose similarity is greater than threshold t will goes to same bucket with probability 1 .



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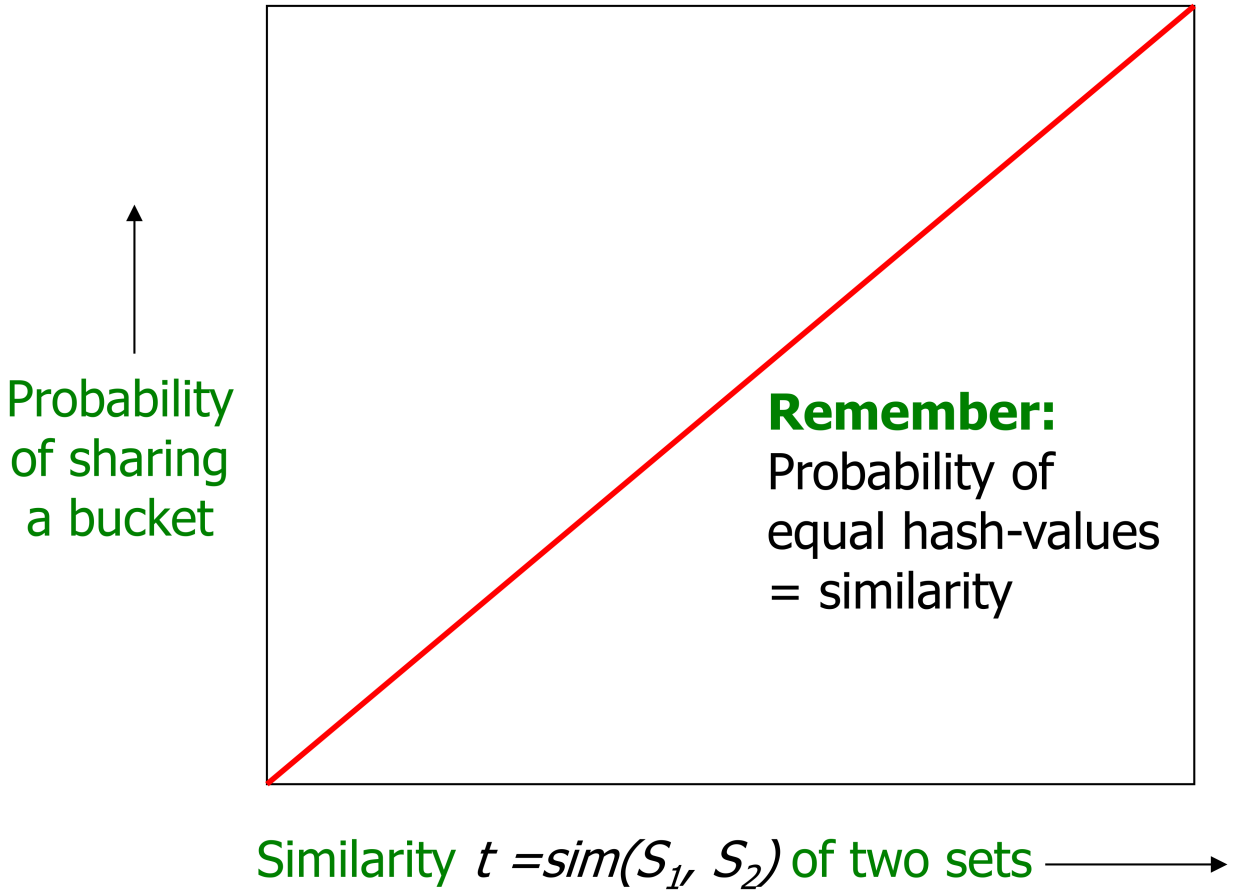
Interpretation of b bands for r rows:

- For $b = 1$ and $r = 1$:



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Interpretation of b bands for r rows:

- For b bands and r rows:

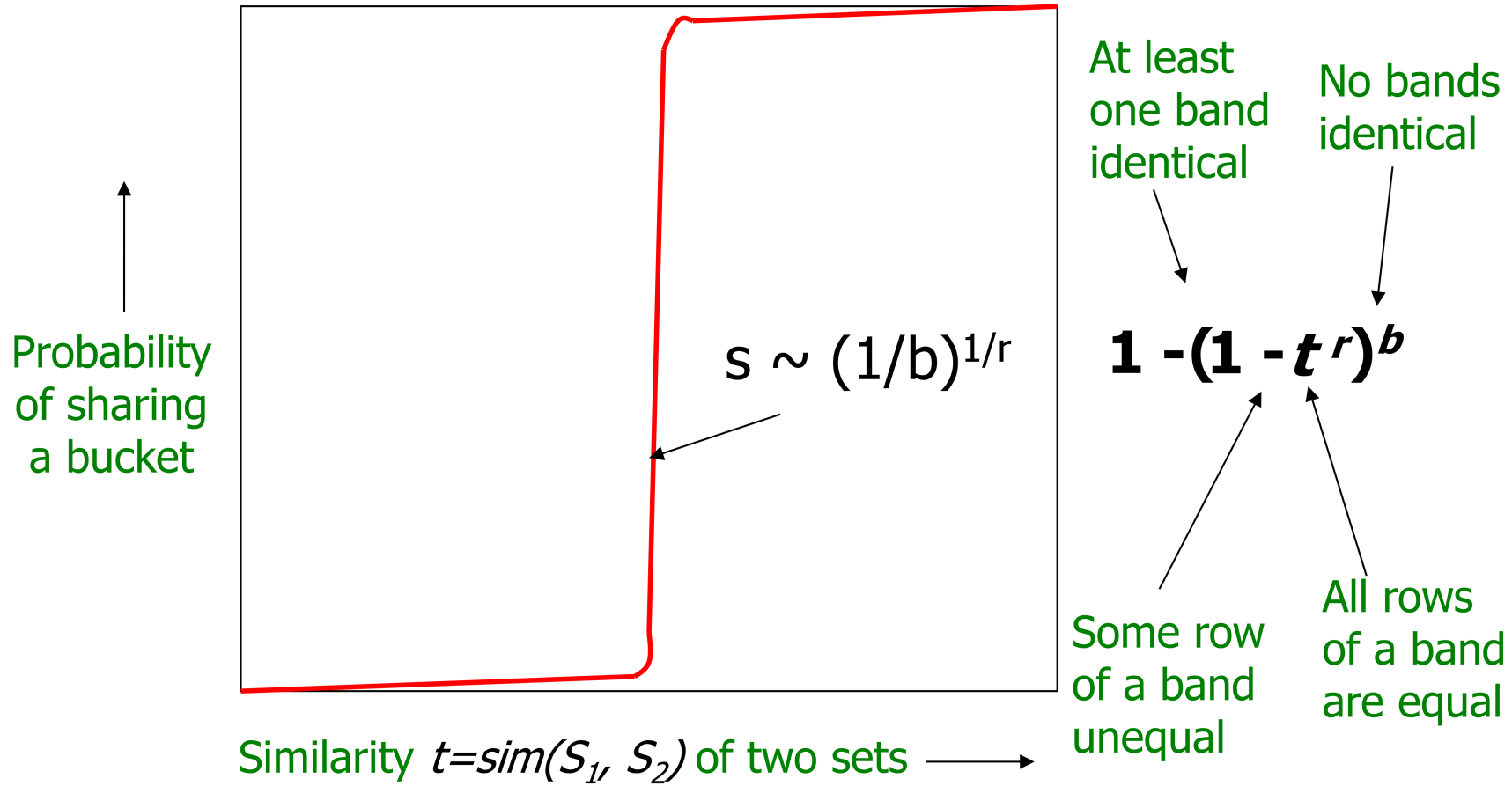

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Table: Actual Truth

Cluster Number	File ID's
1	**[51150, 51240]**
2	**[51270, 51307]**
3	**[53072, 53178]**
4	**[37917, 37950]**
5	**[38216, 38255]**
6	**[38227, 38330]**
7	**[38320, 38432]**
8	**[38438, 38442, 38458]**

Results

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Table: Predicted Duplicates

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2	[37917, 37950]
3	[38354, 38366]
4	[38435, 38457]
5	[53117, 53134]
6	[53174, 53195]
7	[38438, 38442, 38458]
8	[51186, 51210]
9	[38396, 38461]
10	[53072, 53178]

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12	[38217, 38227]
13	[51194, 51277]
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References I

Rajaraman, Anand, and Jeffrey David Ullman. Mining of massive datasets. Cambridge University Press, 2011.

Thank you